Evolutionary Algorithms

Documentation

Traffic Signal Timing Optimisation using PSO, DE, and Hybrid PSO-DE

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Traffic signal timing problem

1. Project Idea

Urban traffic congestion is one of the most significant challenges facing modern cities. Inefficient traffic signal timing leads to increased waiting times, fuel consumption, and carbon emissions. This project addresses the problem of optimising traffic signal timings across a network of intersections using Evolutionary Algorithms, specifically Particle Swarm Optimisation (PSO), Differential Evolution (DE), and a novel Hybrid PSO-DE approach.
The core idea is to find the optimal green-time duration for each traffic light in a simulated urban network such that the total vehicle waiting time is minimised.

2. Problem Formalisation

This problem is formalised as a Continuous Constrained Optimisation problem.
Decision Variables: Green-time durations for each traffic light, represented as a real-valued vector of length equal to the number of intersections.
Objective Function: Minimise the total vehicle waiting time across all intersections over a fixed simulation period of 100 time steps.
Constraints: Each green-time value must lie within the bounds [5, 60] seconds, enforced via clipping (Repair Function approach).
Representation: Each candidate solution is a vector of real-valued green-time durations, one per intersection. A discrete representation was also investigated where timings are rounded to the nearest integer.

3. Main Functionalities

The system provides the following functionalities:
The traffic simulation models a network of three interconnected intersections, each with a traffic light, incoming and outgoing roads, and a vehicle queue. Vehicles arrive according to a Poisson distribution with rush-hour effects between time steps 30 and 70.
The PSO optimiser evolves a swarm of candidate solutions using personal and global best positions. Multiple inertia strategies and topologies are supported.
The DE optimiser evolves a population using mutation, crossover, and selection operators. Multiple mutation strategies, crossover strategies, parent selection methods, and representations are supported.
The Hybrid PSO-DE optimiser combines DE mutation and crossover with PSO memory mechanisms to generate improved candidate solutions.
The experiment runner executes 30 independent runs per configuration, records seeds for reproducibility, and produces statistical summaries and comparison plots.

4. Similar Applications in the Market

Several real-world systems address traffic signal optimisation using similar approaches. SCOOT (Split Cycle Offset Optimisation Technique) is a widely deployed adaptive traffic control system used in cities such as London and Singapore. SCATS (Sydney Coordinated Adaptive Traffic System) is another adaptive system used across Australia and internationally. InSync by Rhythm Engineering uses real-time data and machine learning to optimise signal timings dynamically. VISSIM is a microscopic traffic simulation tool used by traffic engineers to test signal timing strategies before deployment.

5. Literature Review

The following six academic papers are relevant to this project, one per team member:

Paper 1 — for Ghada Saeed : Kennedy, J., & Eberhart, R. (1995). Particle swarm optimization. Proceedings of IEEE International Conference on Neural Networks, 4, 1942–1948. This foundational paper introduces PSO and establishes the velocity update equations used throughout this project.

Paper 2 — for Fatma Badawy : Storn, R., & Price, K. (1997). Differential evolution — a simple and efficient heuristic for global optimization over continuous spaces. Journal of Global Optimization, 11(4), 341–359. This paper introduces the DE algorithm, including the rand/1 mutation strategy and binomial crossover implemented in this project.

Paper 3 — for Jihad Abdel Aziz : García-Nieto, J., Alba, E., Olivera, A. C., & Nebro, A. J. (2009). Optimizing traffic lights with metaheuristics: Reduction of car emissions and consumption. IEEE Congress on Evolutionary Computation, 1–8. This paper applies metaheuristic algorithms including PSO to traffic signal optimisation, directly relevant to the problem addressed in this project.

Paper 4 — for Doha Shafei : Das, S., & Suganthan, P. N. (2011). Differential evolution: A survey of the state-of-the-art. IEEE Transactions on Evolutionary Computation, 15(1), 4–31. This comprehensive survey covers DE variants including best/1 mutation and exponential crossover, both of which are implemented and compared in this project.

5. Literature Review

Paper 5 — for Hajar Ahmed : Zhang, W., & Tan, G. (2011). A hybrid PSO-DE algorithm for numerical optimization. Advances in Swarm Intelligence, 6728, 257–263. This paper proposes a hybrid combining PSO memory with DE operators, directly relevant to the Hybrid PSO-DE approach implemented in this project.

Paper 6 — for Seif Saadani : Lim, W. H., & Isa, N. A. M. (2014). Particle swarm optimization with adaptive time-varying topology connectivity. Applied Soft Computing, 24, 623–642. This paper investigates global versus local topology in PSO, relevant to the topology comparison experiments conducted in this project.

6. Dataset

This project uses a simulated dataset generated by the traffic simulation engine developed as part of the project. The simulation models vehicle arrivals using a Poisson distribution with a base arrival rate of 2 vehicles per time step, doubled during rush hour between time steps 30 and 70. This approach is consistent with established traffic simulation methodologies and allows full control over experimental conditions.
The network consists of three intersections connected by roads with propagation delays of 2 and 3 time steps respectively, representing realistic urban road segments.

7. Algorithm Details and Experimental Results

7.1 Components of PSO

Representation: Each individual is a real-valued vector of green-time durations with values in [5, 60] seconds.
Population: 30 particles per swarm.
Initialisation: Uniform random initialisation within the search bounds.
Fitness Function: Total vehicle waiting time across all intersections over 100 simulation steps. Lower is better.
Velocity Update: Standard PSO velocity equation with inertia weight w, cognitive coefficient c1, and social coefficient c2.
Personal Best: Each particle maintains the best position it has personally discovered.
Global Best: The swarm maintains the best position discovered by any particle.
Termination: Fixed number of iterations (100).
Diversity Preservation: Crowding replacement — when a new candidate is generated, it replaces the most similar existing particle in the swarm only if it achieves a better fitness score. This prevents premature convergence and maintains population diversity.

7.2 PSO Variants Compared

Two inertia weight strategies were compared independently: constant inertia with w=0.5, and linear decreasing inertia from w=0.9 to w=0.4.
Two topology strategies were compared independently: global best topology where every particle is attracted to the single best in the swarm, and local best topology using a ring neighbourhood of size 3.
A constriction factor variant was also implemented and compared.

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7.3 Components of DE

Representation 1 — Continuous: Real-valued green-time durations.

Representation 2 — Discrete: Integer-valued green-time durations, obtained by rounding continuous values.

Initialisation 1 — Random: Uniform random initialisation within bounds.

Initialisation 2 — Uniform: Solutions distributed evenly across the search space.

Parent Selection 1 — Random: Three donor vectors selected uniformly at random.

Parent Selection 2 — Tournament: Three donor vectors selected via tournament selection of size 3, favouring better solutions.

Mutation Strategy 1 — rand/1: Mutant = a + F × (b − c), where a, b, c are random individuals.

Mutation Strategy 2 — best/1: Mutant = best + F × (a − b), where best is the current global best individual.

Crossover Strategy 1 — Binomial: Each parameter independently crossed with probability CR.

Crossover Strategy 2 — Exponential: A contiguous block of parameters crossed starting from a random index.

Selection: Greedy one-to-one selection — the trial vector replaces the target only if it achieves equal or better fitness.

Termination: Fixed number of iterations (100).

7.4 Hybrid PSO-DE

The hybrid algorithm combines the memory mechanism of PSO with the mutation and crossover operators of DE. For each particle at each iteration, DE mutation generates a mutant vector from three random particles, DE crossover mixes the mutant with the current position to produce a trial vector, PSO guidance pulls the trial vector towards the personal best and global best positions using the inertia weight w, and greedy selection retains the guided solution only if it improves upon the current position.
This design exploits DE's ability to explore diverse regions of the search space while using PSO's memory to guide solutions towards historically good regions.

7.5 Experimental Results

All experiments were conducted over 30 independent runs per configuration. Seeds were recorded and stored in results/seeds.json for full reproducibility.

PSO Results Summary:

The best performing PSO configuration was Linear Inertia with Local Topology, achieving a mean score of 19400.2 with standard deviation 547.1 and best score of 18410. This outperformed constant inertia configurations, indicating that adaptive inertia weight improves convergence. Local topology outperformed global topology on average, suggesting that restricted information sharing helps maintain diversity and avoid premature convergence.

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DE Results Summary:

The best single score across all DE configurations was achieved by rand/1 with Exponential Crossover, reaching 17951 in one run. However, this configuration showed higher variance with a mean of 19512, indicating inconsistent performance. The discrete representation with tournament selection was significantly slower due to the computational cost of tournament evaluation, averaging 14 seconds per run compared to under 1 second for continuous variants.

Hybrid PSO-DE Results:

The Hybrid PSO-DE showed competitive performance, sometimes outperforming both PSO and DE individually and sometimes falling between them. This variability is consistent with the literature, which notes that hybrid performance is highly sensitive to parameter settings. With parameters F=0.5, CR=0.7, w=0.3, and 100 iterations, the hybrid achieved scores comparable to the best PSO configurations.

Overall Comparison:

The best average performance across all configurations was achieved by PSO with Linear Inertia and Local Topology. The DE rand/1 Exponential configuration achieved the single best score across all experiments at 17951, demonstrating DE's capacity for exploration. The Hybrid approach showed that combining PSO memory with DE operators is a promising direction, particularly when parameters are carefully tuned.

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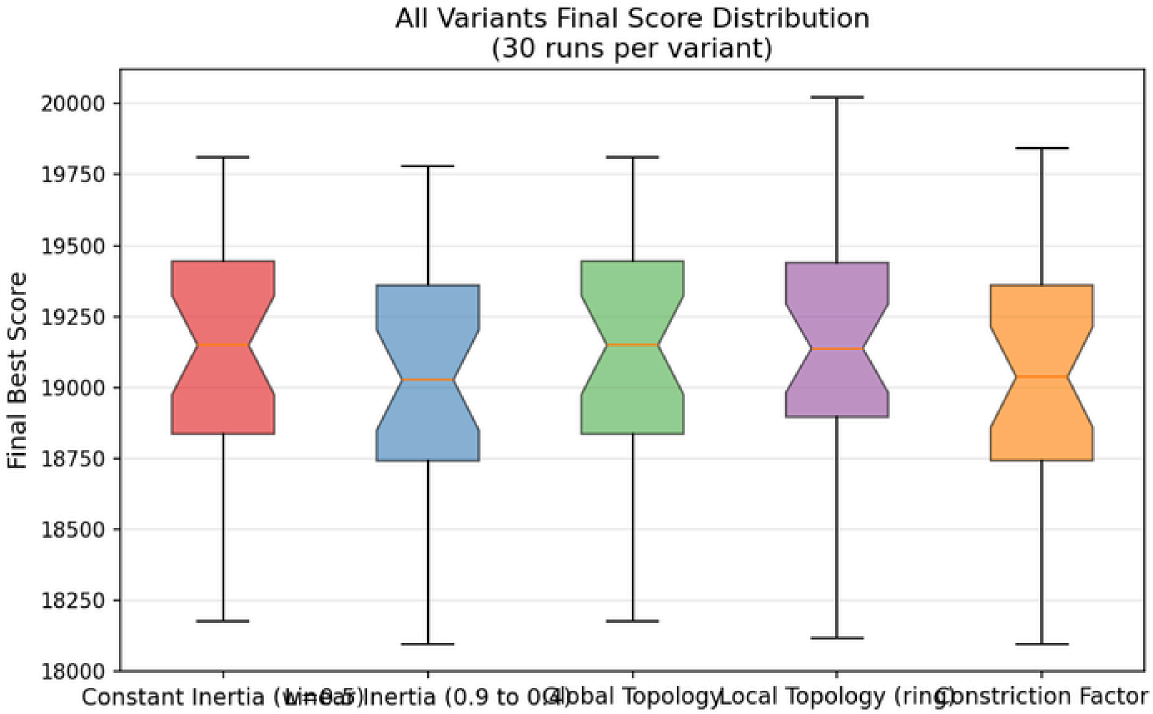
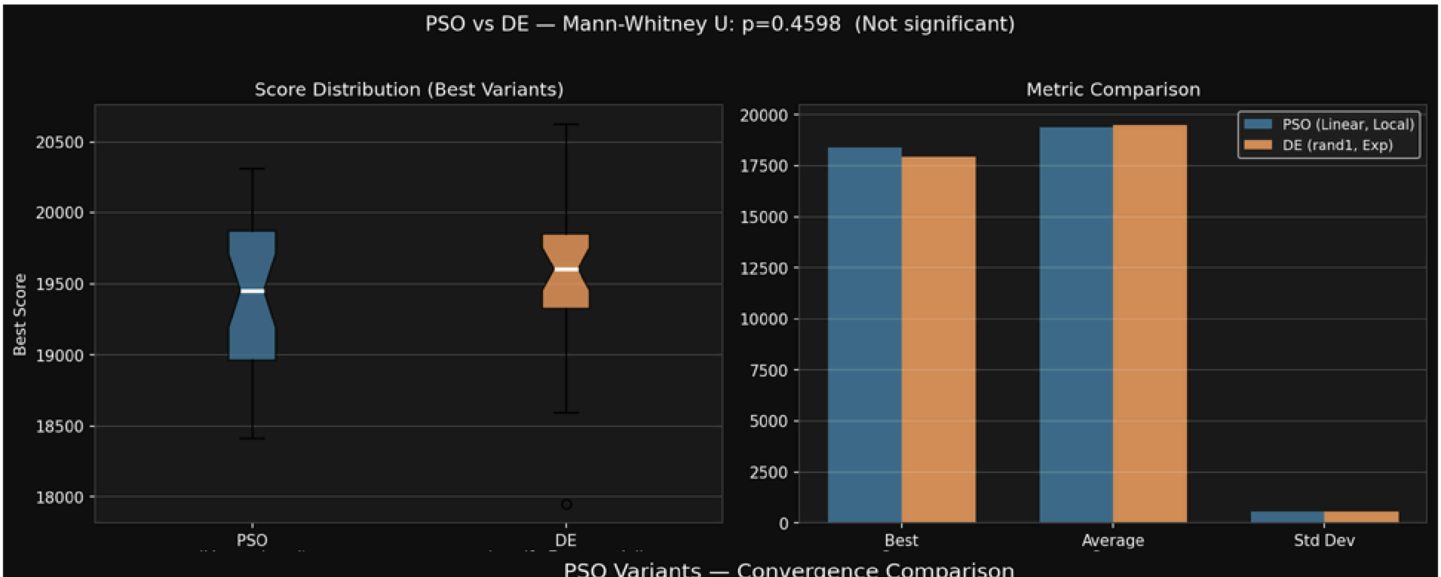
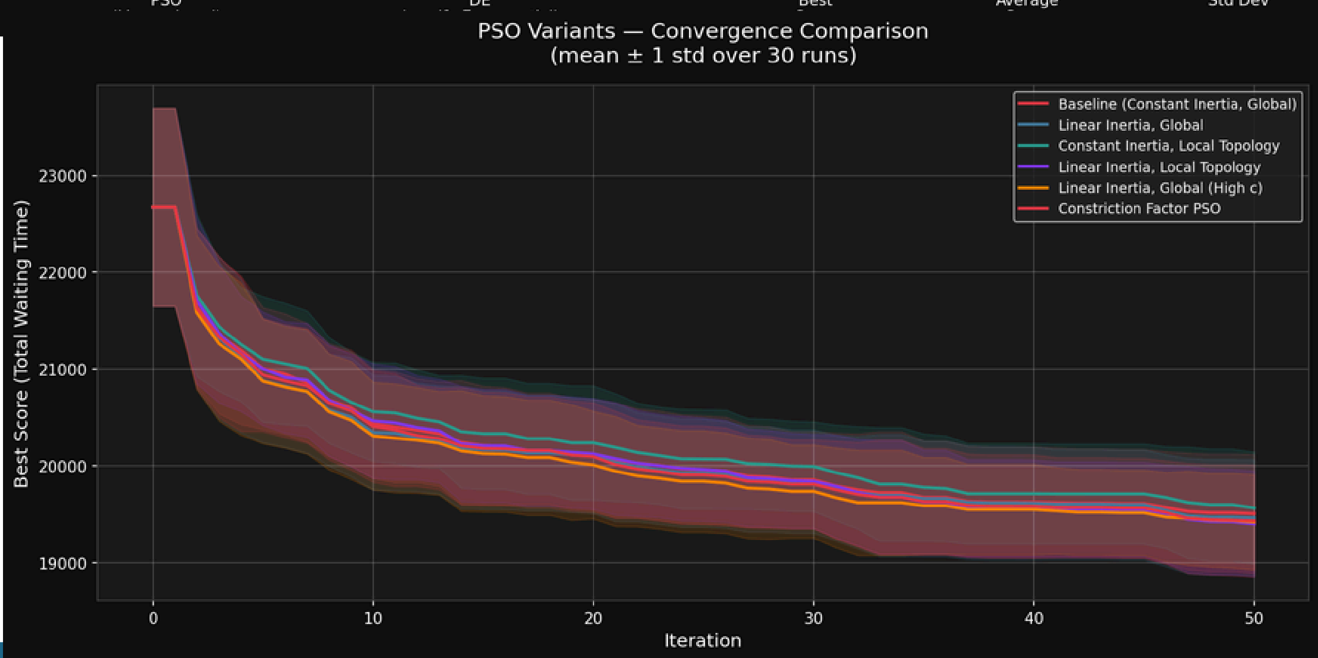
8. Some Results from the code :-







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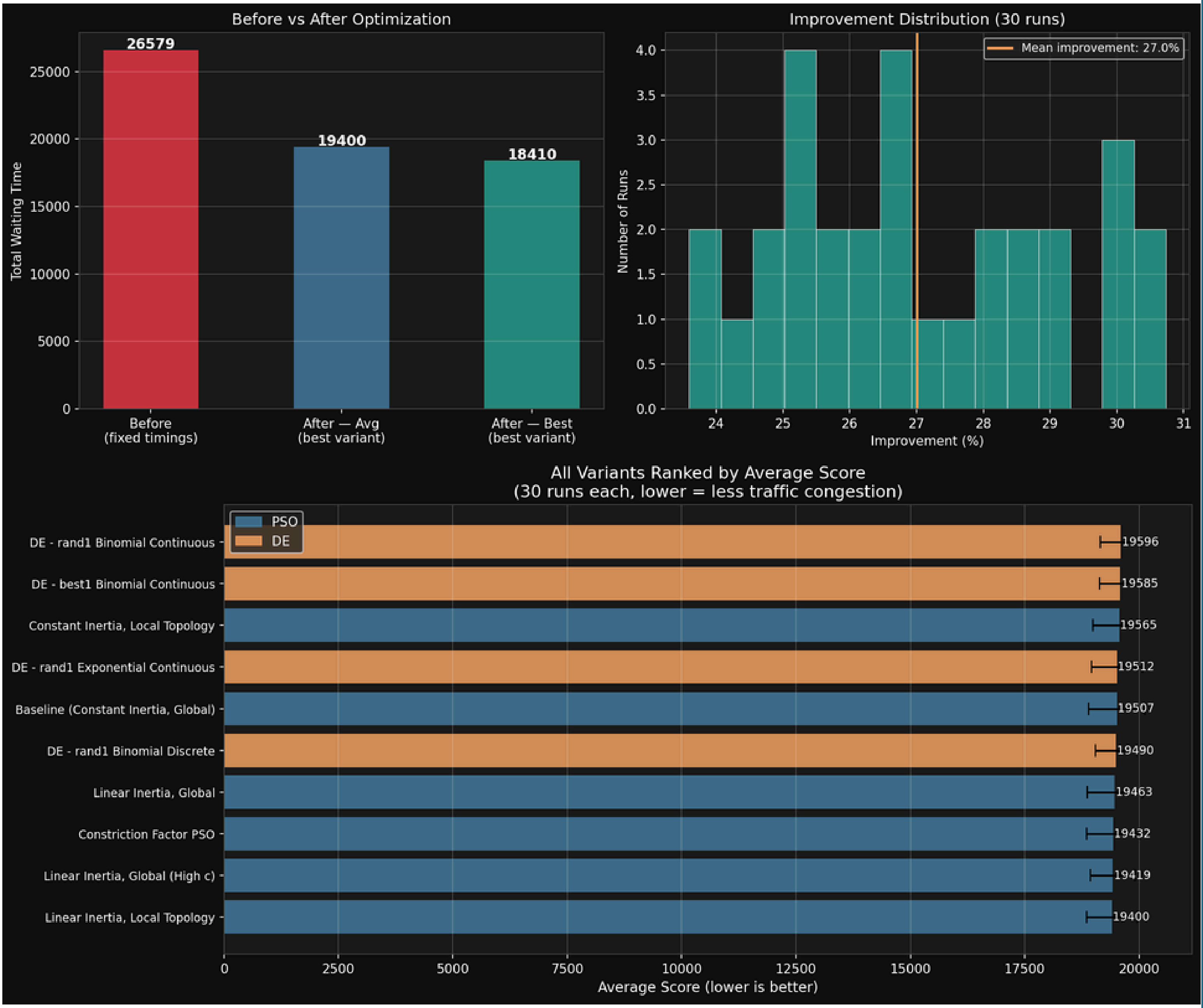

8. Development Platform

Programming Language: Python 3.13

Libraries Used: NumPy for numerical computation, Matplotlib for result visualisation, and the Python standard library for file I/O and statistical analysis.

Project Structure:

- src/simulation — traffic network simulation
- src/pso — PSO algorithm and variants
- src/DE — Differential Evolution algorithm
- src/hypered — Hybrid PSO-DE algorithm
- src/experiments — experiment runner and result storage
- results — output files including JSON, CSV, and seed records

Hardware: Standard consumer laptop. All experiments completed within approximately 10 minutes for the full 30-run suite across all configurations.

Thank You

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